Online Resource 1: Detailed ablations and robustness analyses for calibration-free two-view 3D pose fusion

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1 Purpose

This Online Resource reports prespecified and post hoc ablations, synthetic corruption settings, robustness diagnostics and secondary cohort results for the accompanying article. It preserves the separation between training evidence and evaluation evidence: triangulated poses were resolved only by the evaluation layer and were never used for model training, validation or checkpoint selection.

2 Ablation definitions

3 Complete held-out learned comparison

All A2–A9 outputs were view-order invariant to the reported precision. A3-relative (ROM, peak-speed) retention was A0 (1.156, 1.018), A1 (1.050, 1.654), A2 (1.101, 0.917), A3 (1.000, 1.000), A4 (0.949, 0.828), A5 (0.943, 0.644), A6 (1.000, 1.000), A7 (0.948, 0.821), A8 (1.218, 2.187) and A9 (0.997, 2.615). The A8 and A9 variants demonstrate that relaxing the conservative correction can increase coordinate error and over-amplify peak angular speed.

Table 1 Ablation design. A4–A9 share the same network. Checkmarks indicate active loss groups. A7–A9 are post hoc trunk-twist variants and do not replace the prespecified complete model A6.

ID	Method	Active components
A0	Face stream only	Deterministic single-view output
A1	Side stream in face reference	Deterministic single-view output
A2	Canonical arithmetic mean	Deterministic mean
A3	Deterministic quality mean	Fixed quality weighting
A4	Spatial-objective fusion	Learned residual; recovery, identity and skeletal losses
A5	Rotation-temporal fusion	A4 plus rotation and temporal losses
A6	Complete self-supervised fusion	A5 plus complete-cycle ROM loss
A7	A6 + per-view-peak ROM anchor	A6 plus post hoc trunk-twist anchor
A8	A7 + rotation-parameterized residual	A7 plus post hoc twist residual
A9	A8 + observed twist-rate anchor	A8 plus post hoc rate anchor

Table 2 Held-out comparison ($N = 14$). Mean per-joint position error is reported after one sequence-level similarity alignment to the same-video triangulated pseudo-reference. Differences are method minus A6 with participant-bootstrap 95% confidence intervals and Holm-adjusted paired Wilcoxon p values. Rotation range of motion (ROM) and peak angular speed are A3-relative retention ratios, not comparisons with independent kinematics.

ID	Method	Error (mm)	Difference [95% CI]	p_{Holm}
A0	Face only	77.92 ± 12.52	+17.14 [13.69, 21.34]	0.0011
A1	Side only	75.84 ± 13.93	+15.07 [10.23, 19.48]	0.0018
A2	Arithmetic mean	60.85 ± 8.98	+0.07 [−0.50, 0.59]	1.0000
A3	Quality mean	61.03 ± 8.84	+0.26 [0.17, 0.35]	0.0011
A4	Spatial objectives	62.81 ± 8.86	+2.04 [1.71, 2.36]	0.0011
A5	Rotation/temporal	60.80 ± 8.96	+0.02 [−0.38, 0.43]	1.0000
A6	Complete model	60.78 ± 8.72	–	–
A7	Per-view-peak ROM	61.31 ± 8.87	+0.54 [0.38, 0.69]	0.0011
A8	Twist residual	92.02 ± 9.59	+31.25 [27.78, 34.88]	0.0011
A9	Twist-rate anchor	94.11 ± 9.14	+33.34 [30.00, 36.77]	0.0011

4 Deterministic comparison

5 Synthetic corruptions and robustness

A6 had greater aggregate clean-to-corrupted displacement than A4 and A5 (0.120 versus approximately 0.09 repository units). A temporal-offset sweep over all 137 participants moved A6 by 0.046 and 0.078 units at offsets of ± 2 and ± 4 frames, respectively, essentially matching the deterministic mean. The learned model therefore did not resolve dependence on the upstream global offset.

Table 3 Agreement with the private triangulated pseudo-reference over 137 participants. Units are repository coordinates. The pseudo-reference-fitted row is a leakage diagnostic and is not an eligible label-free method.

Method	Mean	Standard deviation	95% CI of mean
Pseudo-reference-fitted joint weights (leaky)	0.0635	0.0160	[0.0609, 0.0662]
Body-frame average	0.0640	0.0161	[0.0615, 0.0668]
Similarity alignment, stable joints	0.0646	0.0159	[0.0621, 0.0673]
Similarity alignment, body-part weights	0.0649	0.0162	[0.0622, 0.0676]
Similarity alignment, side smoothing	0.0655	0.0159	[0.0629, 0.0682]
Similarity alignment, output smoothing	0.0663	0.0159	[0.0637, 0.0690]
Similarity alignment, all joints	0.0738	0.0195	[0.0707, 0.0771]
World-coordinate average	0.1221	0.0184	[0.1190, 0.1252]
Root alignment and average	0.1222	0.0184	[0.1191, 0.1252]

Table 4 Synthetic-corruption configuration in canonical units.

Corruption	Setting
Independent joint dropout	probability 0.05
Temporal block dropout	joint probability 0.25; 16 frames
Impulse noise	probability 0.02; scale 0.10
Random-walk drift	joint probability 0.10; step scale 0.01
Thorax rotation bias	frame probability 0.10; $\pm 12^\circ$
Frozen segment	joint probability 0.10; 12 frames
Integer time shift	joint probability 0.10; at most 4 frames

6 Additional external comparisons

Two Unity-supervised comparators were evaluated over two direction-held-out folds and three seeds. An exact-extrinsic gate over the two 3D streams averaged 153.645 mm and 41.090°, a 6.74% improvement over its matched direct average. Learnable triangulation averaged 28.058 mm and 14.555°, 1.88% below its matched fixed-triangulation baseline. These rows use Unity native 3D during training and answer a different question from zero-shot private-model transfer.

7 Complete cohort outcomes

All eight cohort-by-cycle-position interactions had $p_{\mathrm{Holm}} = 1.000$. No axial-angle or angular-speed phase cluster survived correction across descriptor families. Secondary phase-localized differences occurred for trunk tilt and wrist wrapping, but are hypothesis-generating and not anatomical validation. Omitting outer-fold fixed effects produced similar coefficients for log angular speed (0.2972) and log jerk (0.4161).

Table 5 Output displacement under the fixed corruption manifest (27 validation participants, one seed). Lower values indicate less movement from each method's own clean output, not recovery toward independent ground truth.

Challenge	A3	A6
Joint dropout	0.0629	0.0873
Temporal block dropout	0.0619	0.0873
Impulse noise	0.0605	0.0605
Random-walk drift	0.0445	0.0445
Thorax rotation bias	0.0312	0.0312
Frozen segment	0.0643	0.0643
Integer time shift	0.0348	0.0348

Table 6 Out-of-fold cohort and repeated-cycle analysis. Cohort effects are elderly-minus-student mixed-model coefficients at normalized cycle position 0.5. MAD effects are participant-level median differences.

Outcome	Cohort effect [95% CI]	p_{Holm}	MAD effect (p_{Holm})
Axial rotation ROM	0.2475 [−0.0242, 0.5192]	0.2966	−0.0016 (1.0000)
Log angular speed (P95)	0.3006 [0.1136, 0.4876]	0.0114	0.0314 (0.4459)
Peak rotation phase	−0.0013 [−0.0568, 0.0542]	1.0000	0.0000 (1.0000)
Trunk tilt (P95)	0.0042 [0.0002, 0.0082]	0.2514	−0.0002 (1.0000)
Wrist wrapping (P95)	0.0289 [−0.0626, 0.1204]	1.0000	−0.0073 (1.0000)
Log cycle duration	−0.0127 [−0.0256, 0.0002]	0.2677	0.0083 (1.0000)
Log dimensionless jerk	0.4233 [0.2336, 0.6130]	0.0001	0.0377 (0.0208)
Log whole-body repeatability	−0.0145 [−0.1055, 0.0765]	1.0000	−0.0003 (1.0000)